

Bicyclist Behavior in the Presence of Two-stage Turn Queue Boxes

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 Show Abstract

Two-stage turn queue boxes are bicycle infrastructure treatments that are implemented to enhance bicyclists' safety and comfort, thereby reducing crash risk at intersections. These boxes provide an alternative for bicyclists to make left turns from right-side cycle tracks or bike lanes and right turns from left-side cycle tracks or bike lanes at multi-lane intersections. Although these solutions are being implemented at various intersections throughout the USA, research on the behavior of bicyclists in the presence of two-stage turn queue boxes at the intersection is very limited. Therefore, it is necessary to investigate how these treatments are used and understand which characteristics may influence their effectiveness. The objective of this study is to create an inventory of two-stage turn queue box design characteristics and investigate bicyclists' utilization of these boxes and their behavior at intersections in the presence of such treatments. A total of 81 two-stage turn queue boxes were identified and their elements were documented. Real-world data with the use of video cameras was then collected from three intersections in Boston, Massachusetts, where a total of eight two-stage turn queue boxes are located. A trajectory analysis was conducted to comprehend the bicyclist's positioning and usage rates of two-stage turn queue boxes for left-turning bicyclists. In contrast with earlier studies, the selected intersections exhibited noncompliance bicyclist behavior in relation to the utilization of the two-stage turn queue box. The findings can serve as a basis for informing design standards and educational initiatives that aim to enhance bicyclist safety at intersections.

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